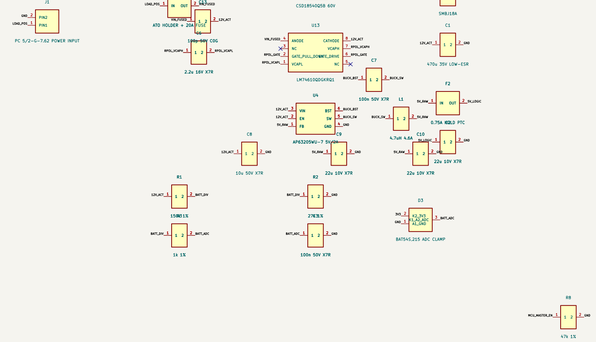
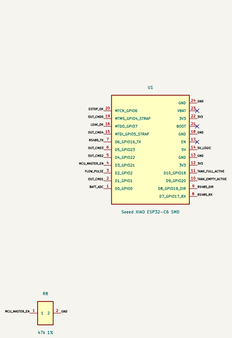
POWER INPUT / 20 A SHARED PATH

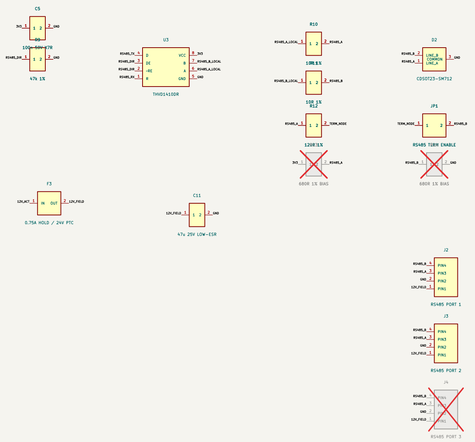
Normal: one 12 V output. Absolute limit: two outputs (<200 W total).



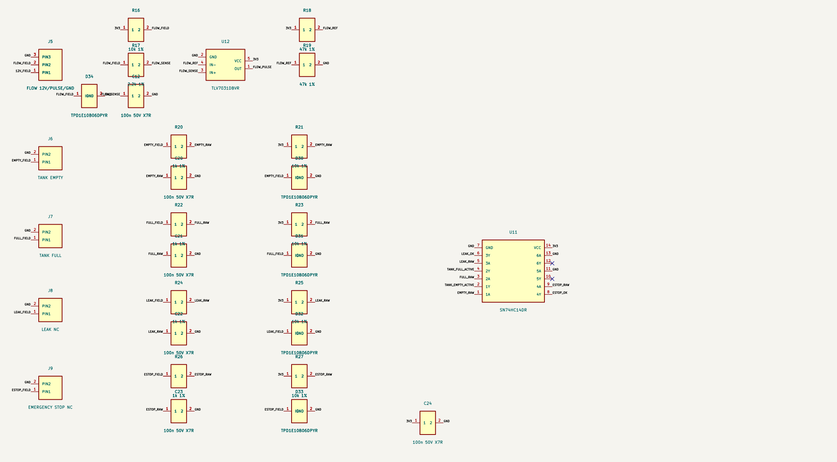
DIRECT-GPIO CONTROLLER / NO GPIO EXPANDER



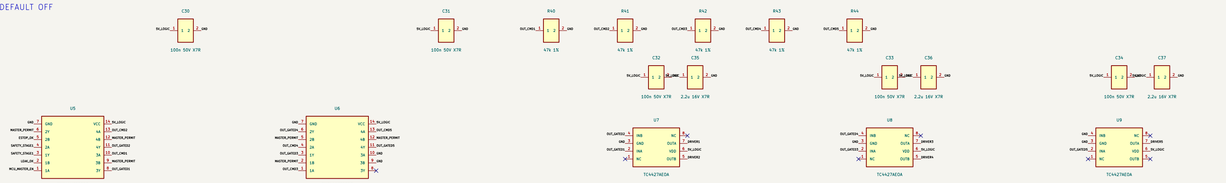
RS485 / GENERIC PARALLEL PORTS / NO SENSOR TYPE ASSUMPTION



FIELD INPUTS / NC SAFETY LOOPS FAIL OPEN



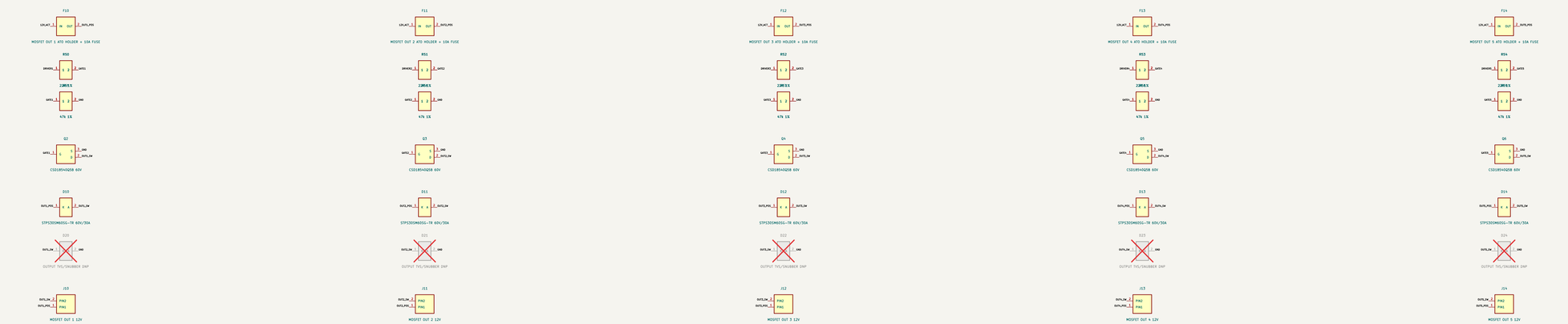
HARD SAFETY GATING / ALL OUTPUTS DEFAULT OFF



Physical outputs are generic. Runtime mapping enforces current FGT A/B/MIXER safety rules.

Never start two motors simultaneously. Third active command is a fault.

12 V MOSFET OUTPUTS / <100 W EACH / 10 A CONTINUOUS TARGET



REV A ENGINEERING BUILD / ORDERABLE PROTOTYPE

Bench validation is mandatory before field use: charger range, inrush, current, thermal rise, faults, and wet-enclosure testing.

JLCPCB PCBA: top-side SMT preferred; connector/fuse THT assembly permitted; verify LCSC stock at release.

ORDERABLE PROTOTYPE — VALIDATE BEFORE FIELD USE
12 V loads <100 W; one normal, two absolute maximum
JLCPCB mixed-technology PCBA baseline

INA Technologies

Sheet: /
File: esp32c6-solar-controller.kicad_sch

Title: FGT ESP32-C6 Solar Fertigation Controller

Size: A1	Date: 2026-08-01	Rev: REV A
KiCad E.D.A. 10.0.5		Id: 1/1